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UTS MTK SAINS DATA

Parameter Personalisasi

1. menentukan N
   $N = 31$
2. menentukan a
   $a = \text{Flor}\left(\frac{31}{10}\right) = 3$
3. menentukan b
   $b = 31 \bmod 10 = 1$

4. hitung k
   $k = a + b + 1$
   $k = 3 + 1 + 1 = 5$

5. hitung θ
   $\theta_0 = (a \times 30) + (b \times 5)$
   $\theta_0 = (3 \times 30) + (1 \times 5) = 90 + 5 = 95^\circ$

6. hitung α
   $\alpha = 0.001 \times (b + 1)$
   $\alpha = 0.001 \times (1 + 1) = 0,002$

$K = 5,\ a = 3 \rightarrow (a+1) = 4$

$\boxed{7}$

$\boxed{7.1}$ Identifikasi pola

(a)

rumus : $S_n = 50 + 5n + 4(-1)^n$

Jwb : $S_1 = 50 + 5(1) - 4 = 51$
$S_2 = 50 + 10 + 4 = 64$
$S_3 = 50 + 15 - 4 = 61$
$S_4 = 50 + 20 + 4 = 74$
$S_5 = 50 + 25 - 4 = 71$
$S_6 = 50 + 30 + 4 = 84$
$S_7 = 50 + 35 - 4 = 81$
$S_8 = 50 + 40 + 4 = 94$
$S_9 = 50 + 45 - 4 = 91$
$S_{10} = 50 + 50 + 4 = 104$
$S_{11} = 50 + 55 - 4 = 101$
$S_{12} = 50 + 60 + 4 = 114$

Pemisah : $S_n = (50 + 5n) + 4(-1)^n$
trend $= 50 + 5n$
fluktasi $= \pm 4$
Osilasi karena tdk aritmetika / geometri

$\boxed{7.2}$

(b) T_1 $\quad T_1 = \Sigma(50 + 5n + 4(-1)^n)$

Pisah
- $\Sigma 50 = 600$
- $\Sigma n = 78 \rightarrow \Sigma 5n = 390$
- $\Sigma(-1)^n n = 0$

$T_1 = 600 + 390 = 990$

T_2

$S_n^2 = (50 + 5n)^2 + 8(50 + 5n)(-1)^n + 16$
$T_2 = 85{,}862$

$\boxed{7.3}$

(d) Deret Geometri
Dik $G_n = 5(1.03)^{n-1}$

total 10 bulan : $S_{10} = \dfrac{5(1.03^{10} - 1)}{0.03} \approx 57{,}32$

Cari n saat > 500

$$\frac{5(1.03^n - 1)}{0.03} > 500$$

$1{,}03^n > 4 \rightarrow n \approx 47$

Dipindai dengan CamScanner